Event-Driven Sentiment Analysis of Indian Audiences Towards Minority Content Creators

Howdo Indianaudiences' sentimentstowards Muslim content creatorschange overtimefollowing major geopolitical events?

1. Introduction

Social media platforms shape how people engage with content creators. In India public sentiment can sharply shift following geopolitical events like Phalgam attack.

This study aims to study how Indian audience sentiment towards muslim content creators change before and after such events.

2.Hypothesis

Major geopolitical events in India lead to a significant shift in audience sentiment toward Muslim content creators, with a measurable increase in negative sentiment in the post-event period compared to the pre-event period.

3.Methodology

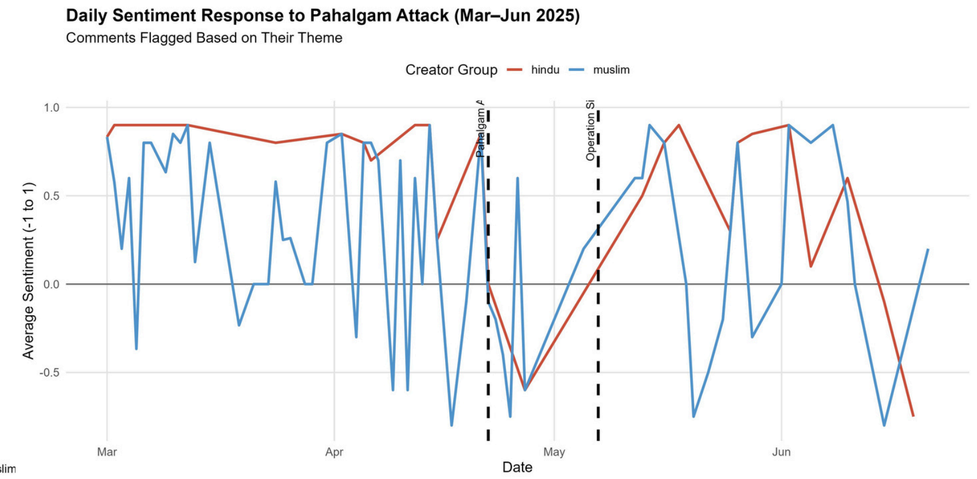
DataCollection→Cleaning→Classification → Comparative Analysis

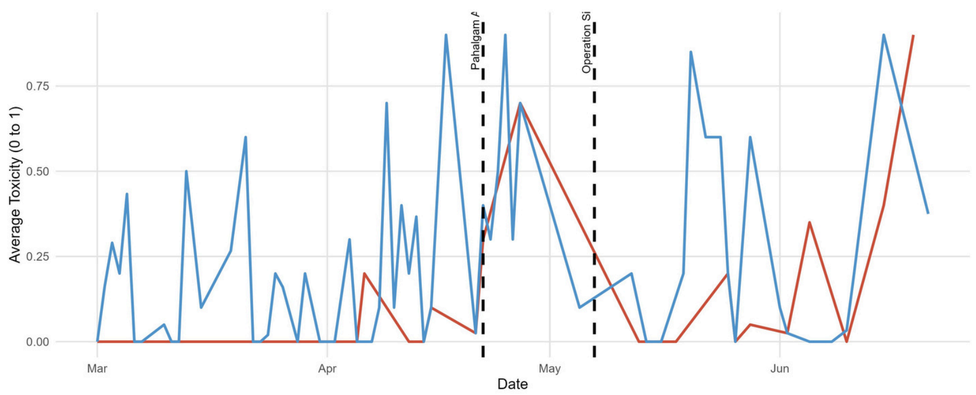
- A randomised sample of 13 YouTube creators (7 Muslim, 6 Hindu) with varied non-political content was selected; comments and replies from videos (March–June 2025) were collected using API via google-api-python-client
- The dataset was compiled and structured using Pandas; ~150 comments per creator were sampled, resulting in ~700 (Muslim) and ~600 (Hindu) comments for analysis
- Text was preprocessed (noise removal, timestamp standardisation) to enable temporal segmentation around key events

Analysis was implemented in Google Colab using Python, applying lexicon-based (sentiment) and rule-based (toxicity, themes) NLP techniques
Sentiment classification was performed using VADER, alongside toxicity detection and thematic categorisation
The processed data was analysed to identify pre–post event trends and comparative differences between creator groups

4.Analysis

- Sentiment and toxicity towards Muslims shows high volatility when comments are flagged based on nationalist, political or religious themes.
- There are spikes in toxicity for both Hindu and Muslim creator comment streams, indicating strong short-term emotional shocks.





Hindu series shows more sustained sentiment decline, while Muslim series shows higher short-term volatility in both sentiment and toxicity.
Periods of falling sentiment consistently align with rising toxicity, indicating increased polarization and hostility during and after key political events.
Neither sentiment nor toxicity returns to pre-March baseline levels, suggesting a lasting change in discourse rather than a temporary reaction.

6.Caveats, Limitations

- Based mainly on YouTube comments and a limited set of creators, so it may not represent the full audience
- LLM-based methods may not fully capture sarcasm or other colloqial phrases
- Analysisshowstrends,not strong causal relationships

7.Future work

- Expand tomore platforms and a larger sample of creators
- Improve handlingof sarcasm and nuanced language
- Extend analysis over a longer time period
- Use better multilingual models to improve accuracy

8.Conclusion

- External events cause short term volatility in sentiments and toxicity
- Volatility is more stark when focusing on comments containing themes of nationalism, religion and politics. Volatility in religion seems more long-term.
- Audience reactions toward Muslim creators are more variable and context-dependent compared to Hindu creators.

DDD
Elements of Statistical Machine Learning & Politics of Data

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